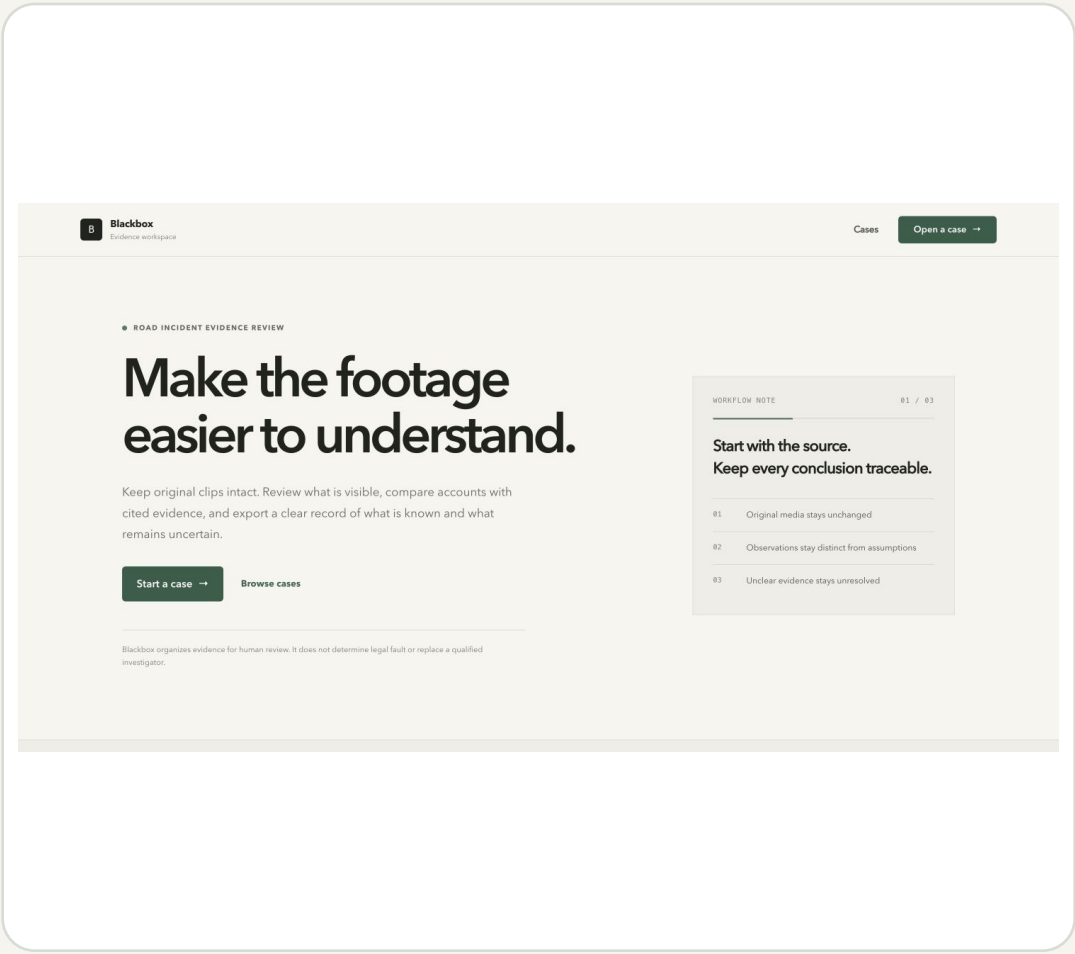
Blackbox

Road-incident evidence, organized for human review.

Keep the source intact. Make the review clearer.

A careful evidence workspace

Not an automated fault verdict.



Current product interface

THE PROBLEM

Footage exists. A usable account often does not.

After a crash, evidence is scattered across phones, dashcams, CCTV, and recollections. People rarely have an easy way to preserve it, find the decisive moments, or show an expert what is known and what is still uncertain.

1.19 million

estimated road-traffic deaths per year

worldwide in WHO’s 2023 report

172,890

road deaths recorded in India in 2023

THE EVIDENCE GAP

- **Preservation**
  Original clips and their context can be lost or altered.
- **Interpretation**
  A polished summary can hide uncertainty and missing context.

Sources: WHO, Global status report on road safety 2023; Ministry of Road Transport and Highways, Road Accidents in India 2023.

THE PRODUCT

A review workspace, not a verdict machine

Blackbox helps a person move from raw road-incident media to a structured, source-linked review package.

01

Preserve

Record the source file, integrity hash, and processing history.

02

Review

Inspect frames, tracks, statements, and what the evidence cannot establish.

03

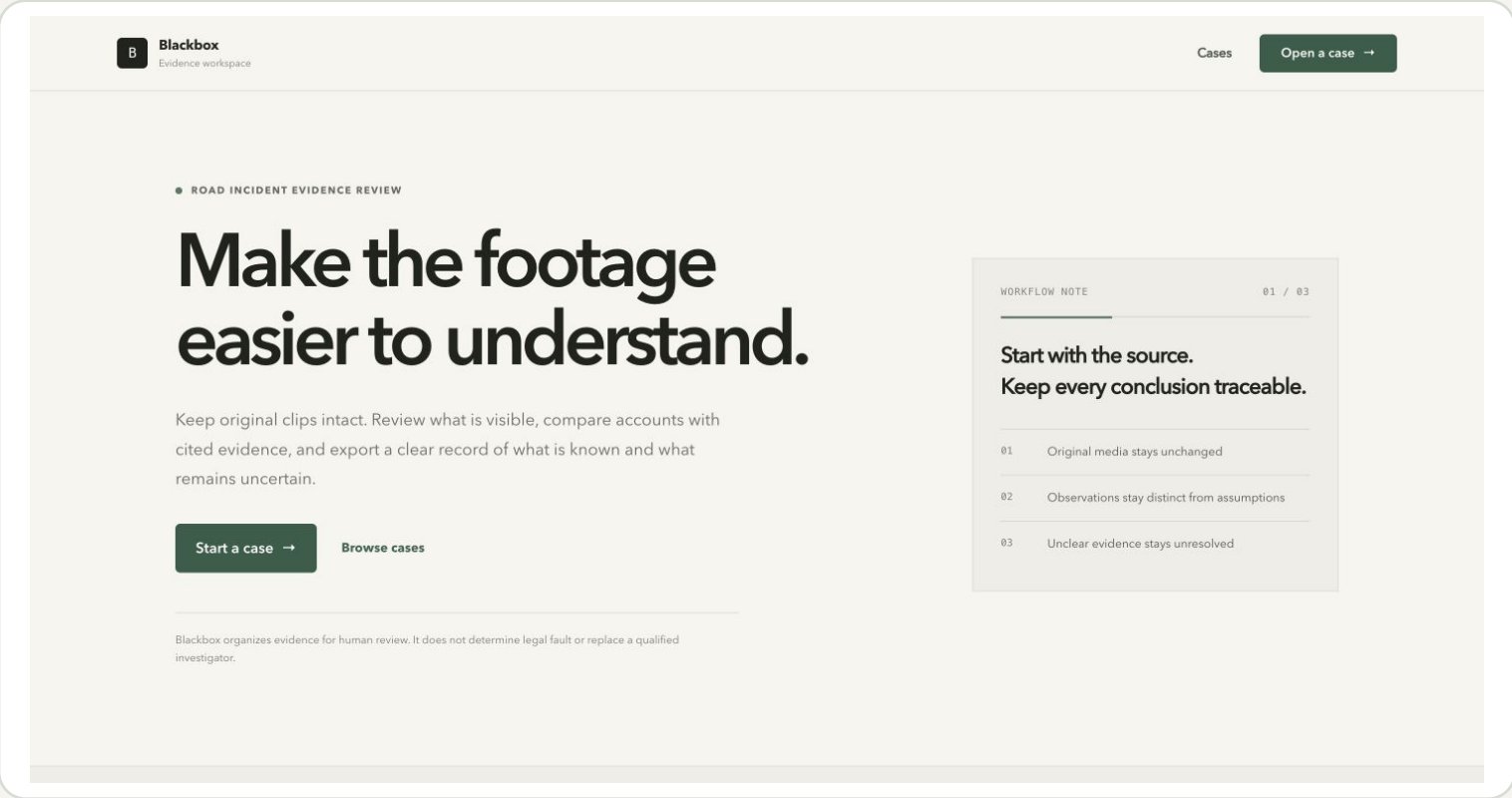
Share

Export a clear record of observations, assumptions, and open questions.

Designed for people who need an understandable first review and for professionals who need traceable inputs.

PRODUCT SCREEN

A calm interface for a high-stress problem





Evidence handling principles

The current app emphasizes source integrity, review, and clear limits.

A PRACTICAL ENTRY POINT

One clip can help. It cannot answer everything.

ONE CLIP CAN SUPPORT

- **A preserved source**
  File hash, metadata, and a record of later processing.
- **A reviewable timeline**
  Visible events and reviewer notes tied to moments in the clip.
- **A bounded claim check**
  Where supported, test a narrow observable claim; otherwise keep it unresolved.

ONE CLIP CANNOT MAGICALLY REVEAL

- **A second viewpoint**
  Occluded actions remain unseen.
- **Metric speed without scale**
  Perspective and camera motion may prevent physical measurements.
- **Legal responsibility**
  That requires full context and qualified human judgment.

Blackbox should return useful, limited findings from one clip and make the missing evidence visible.

HOW IT WORKS

A traceable path from upload to review

STEP 01

Source media

Original file + consent

STEP 02

Processing

Metadata, frames, tracks

STEP 03

Review

Claims + measurements

STEP 04

Report

Sources + limitations

Meta model integrations are optional and consent-gated.

- **SAM 3.1**
  Optional video tracking adapter; the local motion baseline remains available.
- **Muse Voice**
  Optional transcription adapter for recorded statements.

The current claim extraction path uses application code and reviewed evidence; it does not rely on Spark as a physics authority.

The product must show how much it does not know

Evidence quality is part of the output, not a footnote.

OBSERVED	Directly visible in a source clip, with a frame or time reference.
MEASURED	Calculated from reviewed data and declared geometry.
ASSUMED	An input that the footage cannot establish on its own.
SIMULATED	A conditional scenario, never a historical fact.
UNKNOWN	A question the available evidence cannot resolve.

No legal fault verdict. No authenticity claim from a hash. No speed estimate without adequate calibration.

PRODUCT WEDGE

From “what is visible?” to “what evidence is missing?”

The differentiator is the evidence workflow around the models.

Evidence Rescue Radar

Rank missing evidence by relevance and urgency; prepare a preservation-request draft.

Rival hypotheses

Expose more than one explanation when the observed tracks allow it.

Identifiability certificates

Say whether a value is measurable, conditional, or not recoverable.

Prevention view

Aggregate consented cases only after enough reviewed data exists.

These are prototype capabilities and design targets; their accuracy and operational value still require independent evaluation.

USERS AND ADOPTION

Start with review. Earn the right to advise.

01

First users

People affected by crashes who have a clip; legal-aid teams and advocates who need to triage evidence.

02

Professional reviewers

Collision investigators and claims reviewers who can inspect provenance and point out analytical failure.

03

Longer-term partners

Road-safety organizations and local agencies that may use consented, aggregated evidence for prevention research.

Commercial direction to validate

Free or low-friction intake for individuals; paid review tools or deployments for organizations. This is a business hypothesis, not proven demand.

POSITIONING

The gap is between raw video and expert review

Raw clips

Easy to collect; difficult to organize and review.

Blackbox

Source-linked triage with explicit limitations.

Expert analysis

Required for authoritative forensic conclusions.

Positioning hypothesis

Blackbox complements professional investigation; it does not replace established forensic tools or qualified experts.

The hackathon claim is the integrated, accessible workflow, not that video reconstruction is new.

WHAT MUST BE PROVEN

Measure reliability before expanding the promise

01	**Ground truth**	Record staged, low-speed scenes with known distances and independent timing.
02	**Robustness**	Report failures across viewpoint, occlusion, compression, and frame-rate variation.
03	**Calibration**	Publish reprojection error and withhold metric results when fit is poor.
04	**Human review**	Ask independent reviewers what they can and cannot conclude from the export.
05	**Reproducibility**	Record inputs, parameters, versions, and seeds for numerical results.

Current status: code paths and interface exist; no broad accuracy claim is made in this pitch.

BLACKBOX

Keep the source intact.
Keep the uncertainty visible.

The ask

Help us validate the workflow with staged footage,
collision-review expertise, and a responsible pilot partner.

A source-linked first review for road incidents.

WHO 2023: 1.19M estimated annual deaths. India MoRTH 2023: 172,890 deaths.

WHO: who.int/publications/i/item/9789240086517 | MoRTH: morth.nic.in/sites/default/files/Road-Accident-in-India-2023-Publications.pdf